TEMPORAL WEIGHTING AND MULTI-REPRESENTATION FUSION FRAMEWORK FOR AUDIO-BASED INTELLIGIBILITY PREDICTION

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ABSTRACT

Predicting lyrics intelligibility for people with hearing loss is quite challenging because singing introduces several factors that differ from normal speech. Singers use different vocal styles and articulation, the voice is often mixed with heavy background music, and the listener's hearing ability further affects how well the lyrics can be understood. We proposed an ensemble framework that brings together linguistic cues, deep audio representations that help understand the transcript, and handcrafted audio features that capture the difficulty of recognizing lyrics in the presence of music. The framework contains two models. The first model uses a frozen Whisper encoder and learns 1500 temporal weights to highlight the frames that are most important for predicting lyrics intelligibility. This model indirectly mimics how people with reduced hearing ability focus on certain parts of the signal. The second model uses Whisper and HuBERT embeddings combined with handcrafted audio features and applies a traditional regression approach to predict the intelligibility score. The main idea behind choosing this architecture is to leverage the strong linguistic and transcript-level signals from foundation models while also learning hearing-loss-related patterns through temporal weighting. The resulting system is computationally efficient, interpretable, and suitable for small-sample settings common in lyric intelligibility datasets.

Index Terms— Lyric intelligibility, hearing loss, audio representation learning, temporal weighting, ensemble modeling.

1. INTRODUCTION

Music plays an important role in human well-being and enjoyment, and understanding the lyrics is a vital part of the music experience. However, for people with hearing loss, understanding lyrics clearly is a significant challenge. Even though speech intelligibility prediction is a well-established field with metrics such as STOI, these metrics cannot be directly applied to lyric intelligibility, as they fail to provide an accurate picture in the presence of complex musical mixtures and wide variability in vocal style and articulation (1). Developing a robust model to predict lyric intelligibility is therefore a necessary step toward creating next-generation hearing aids and innovations in audio processing algorithms that can enhance music perception for all listeners.

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2. DATASET CHARACTERISTICS

The training data comprise a total of 8,802 data points, each with a simulated listener-reported intelligibility score ranging from perfect understanding to no understanding at all. The data show a bimodal distribution, with one third of the samples at the extreme values of either 0 or 1. Among these, 0 occurs twice as often as 1, indicating that speech degradation in noisy environments more often results in complete lack of understanding rather than gradual impairment. Hearing loss severity is almost evenly stratified across the three groups. As expected, the no-loss group has substantially higher intelligibility scores (mean 0.58) than the other two loss groups. Whisper ASR does not always match how humans perceive lyrics. Even when Whisper gives a perfect transcription (correctness = 1.0) on 942 samples, human intelligibility is only about 0.75 on average, and in 18.2% of those cases listeners score below 50%. The gap is also visible when Whisper completely fails (correctness = 0.0). This shows that people use context, prior knowledge, and inference to understand speech in difficult conditions, while an ASR system cannot.

3. METHODOLOGY

The overall methodology is depicted in Figure 1 and is motivated by two main goals: (i) to exploit the frame-level linguistic and phonetic information encoded by the foundation model, and (ii) to explicitly model the patterns that matter for listeners with reduced hearing.

3.1. Model A: Whisper Temporal-Weighting Regressor

Encoder features: We use a frozen Whisper encoder that outputs a 1500×1280 framewise embedding for the input audio.

Temporal weights head: A lightweight temporal weighting vector $\mathbf{w} \in \mathbf{R}^{1500}$ is learned (implemented as a small feed-forward layer producing 1500 values). A sigmoid activation is applied to constrain the weights to the interval $(0, 1)$. The weighted representation is then computed as

$$y_A = \sigma\left(\mathrm{FF}\left(\sum_{t=1}^{1500} w_t \cdot e_t\right)\right), \tag{1}$$

where e_t denotes the encoder frame at timestep t. The total number of learned parameters is very small (2781 in total), which is appropriate given the limited sample size, providing roughly four samples per parameter (2). The intuition behind this design is similar to an attention mechanism that identifies and emphasizes the most informative regions. The learned weights effectively mimic selective temporal amplification, highlighting syllables or segments that contribute most to predicting the intelligibility score.

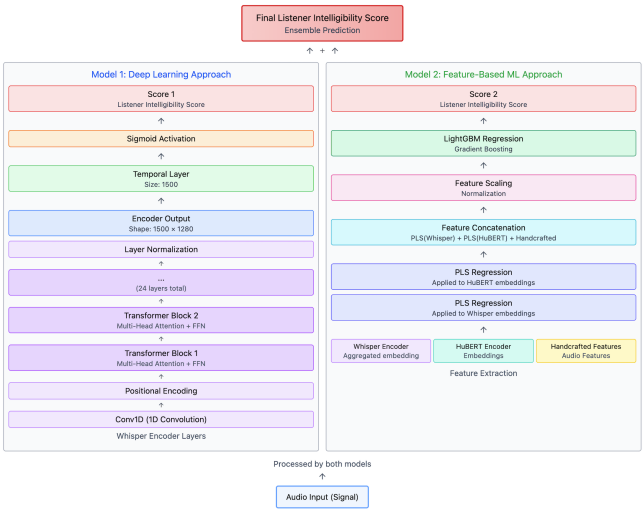

Fig. 1. Block diagram of the proposed ensemble architecture

3.2. Model B: PLS + LightGBM Regressor

Features: This model uses three types of audio-based representations: (i) aggregated Whisper encoder embeddings capturing transcript-related cues, (ii) HuBERT pooled embeddings that provide acoustic and linguistic information, and (iii) handcrafted features such as STOI (vocal vs. source-separated), SNR, Whisper correctness (Whisper transcript prediction vs. prompt), rarity score of the predicted words, high-frequency energy ratio, metadata attributes, and onset rate. These features are included to capture the factors that influence lyric intelligibility for individuals with listening disabilities as well as general music listeners. The Whisper and HuBERT embeddings undergo dimensionality reduction through Partial Least Squares (PLS), retaining components that maximize covariance with the intelligibility score.

Training: A LightGBM regressor is trained on the concatenated PLS components and handcrafted features to predict the intelligibility score. The model is tuned using careful cross-validation to ensure robustness under the small-sample setting.

3.3. Ensemble and Calibration

Weight selection: The final prediction is obtained by a weighted average of the two predicted outputs. The weight for each model is selected through a grid search on the validation set. For a weight pair $(\alpha, 1-\alpha)$, the ensemble score is computed as

$$\hat{y} = \alpha\, y_A + (1-\alpha)\, y_B, \tag{2}$$

and the value of α that yields the lowest validation error is chosen.

4. RESULTS AND DISCUSSION

The dataset was split into training and test partitions using an 80–20 ratio. The performance gap between the train and test sets for both models is small, and the ensemble produced the most consistent performance overall. The correlation and RMSE values for the train, validation, and test sets (validation values obtained from the Cadenza website) are reported in Table 1. These results indicate that the temporal-weighting mechanism is effective at identifying key acoustic regions that influence lyric clarity, while the PLS-based regression model benefits from combining transcript, acoustic, and perceptual features. All experiments were conducted on a workstation with 32 GB RAM. Model A required approximately 24 minutes of training time on an NVIDIA GPU, while Model B completed training in about 3 minutes on an 8-core CPU.

Table 1. Performance of the proposed Ensemble Architecture

Split	Correlation	RMSE
Train	0.7613	24.01
Test	0.7542	24.32
Validation	0.6780	26.69

References

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